# ACC3018 Applied Analytics Capstone

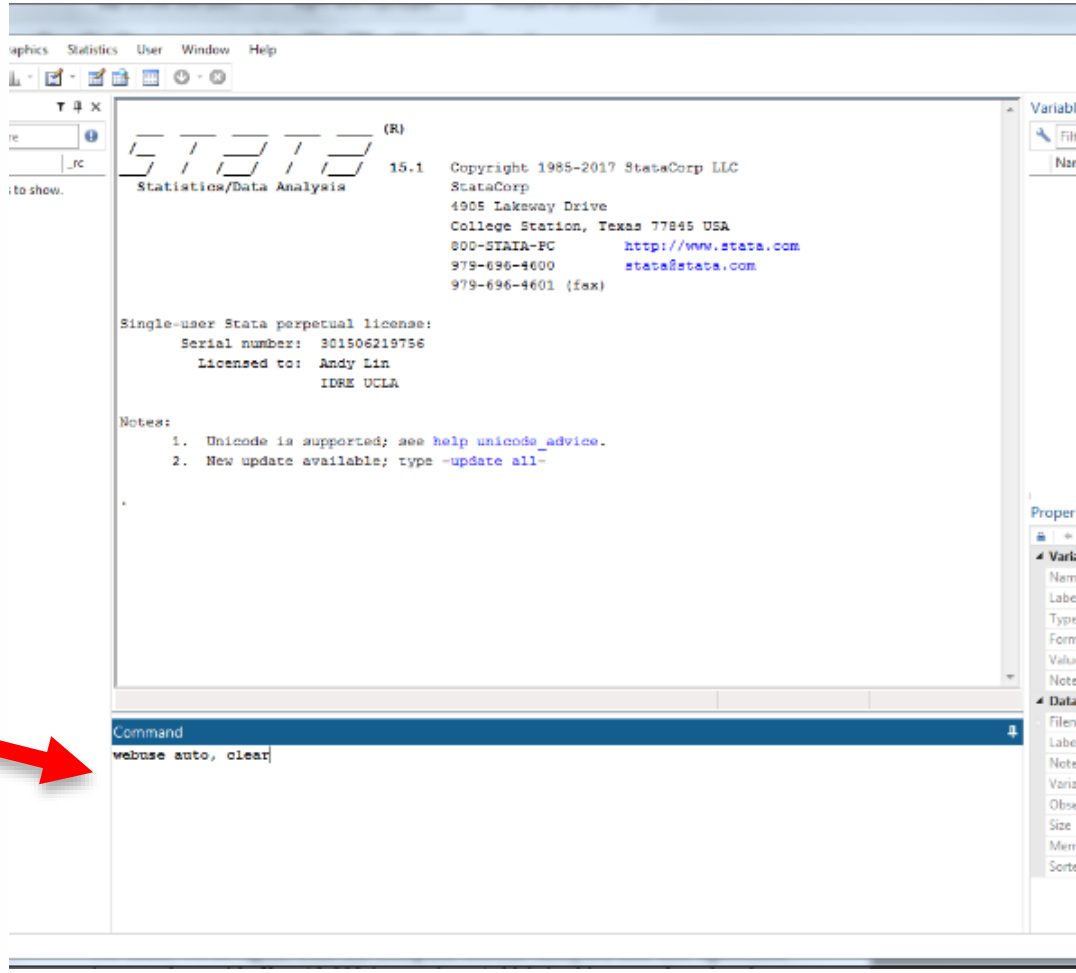
## Seminar 5

# Agenda

- Introduction to STATA (full content from: https://stats.oarc.ucla.edu/stat/data/intro_stata_class/introduction%20stata%20for%20classes.pptx)
- Summary
- Correlation matrix
- Regression results
- Robustness checks

# Introduction

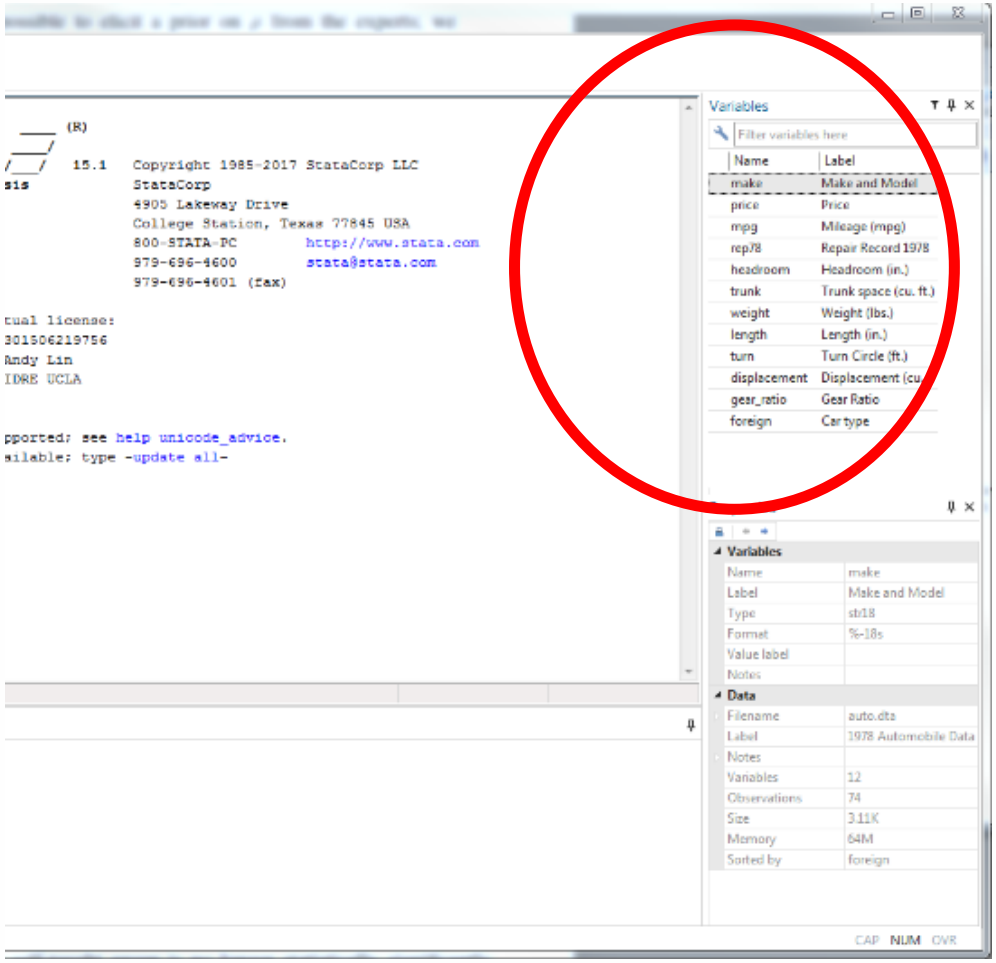
You can enter commands directly into the Command window



# Variables window

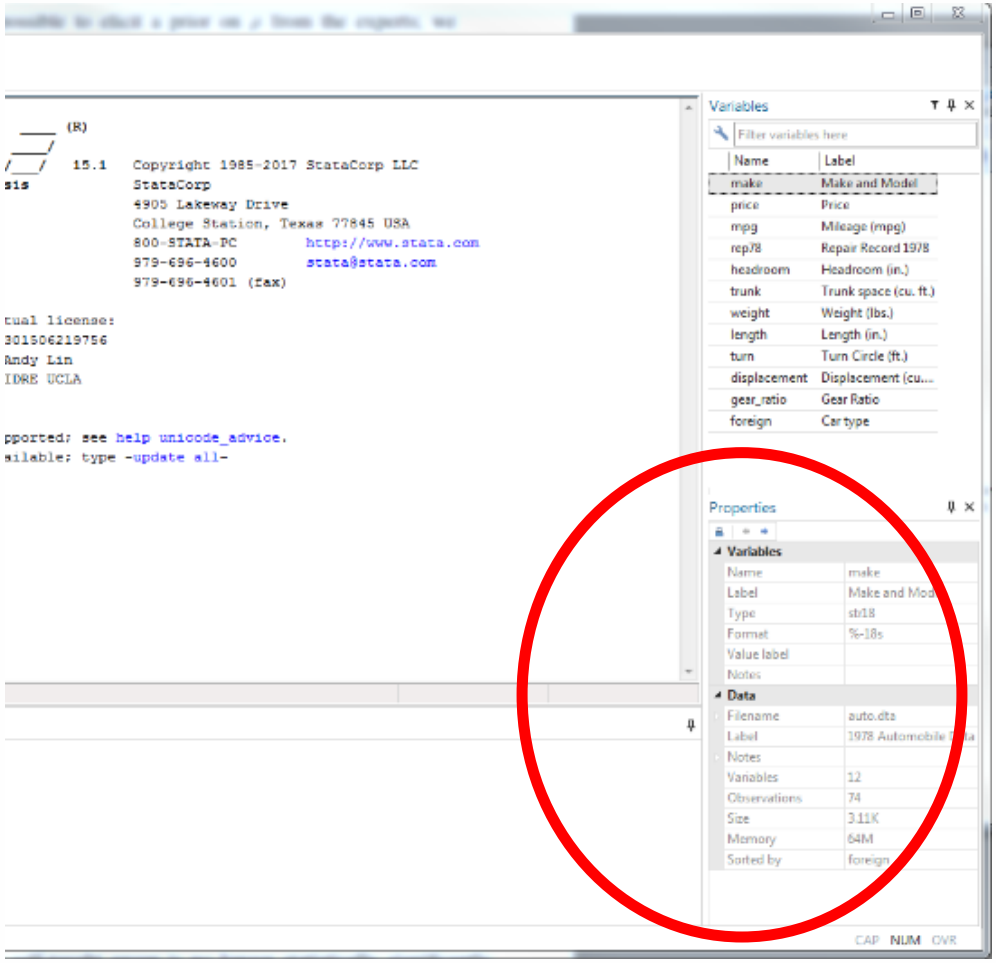
Clicking on a variable name will cause its description to appear in the Properties Window

Double-clicking on a variable name will cause it to appear in the Command Window



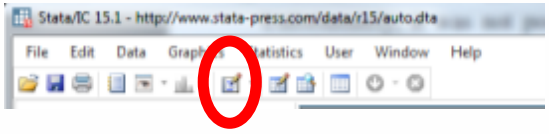
# Properties window

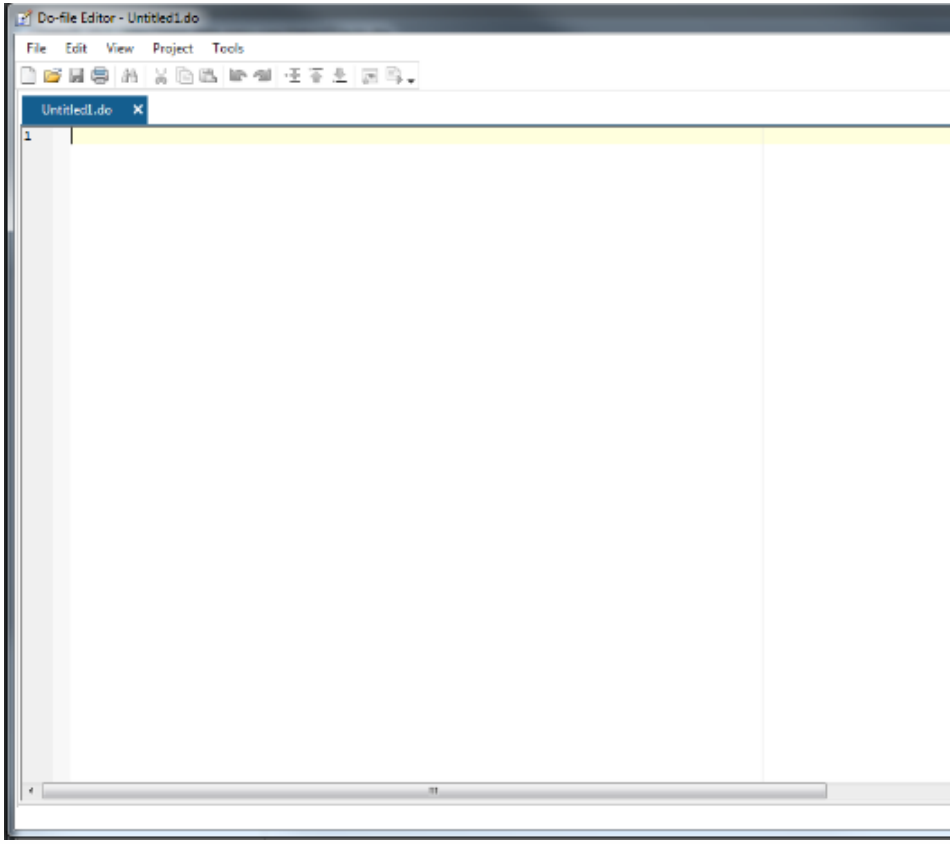
The **Variables** section lists information about selected variable
The **Data** section lists information about the entire dataset



# Opening the do-file editor

Click on the pencil and paper icon on the toolbar



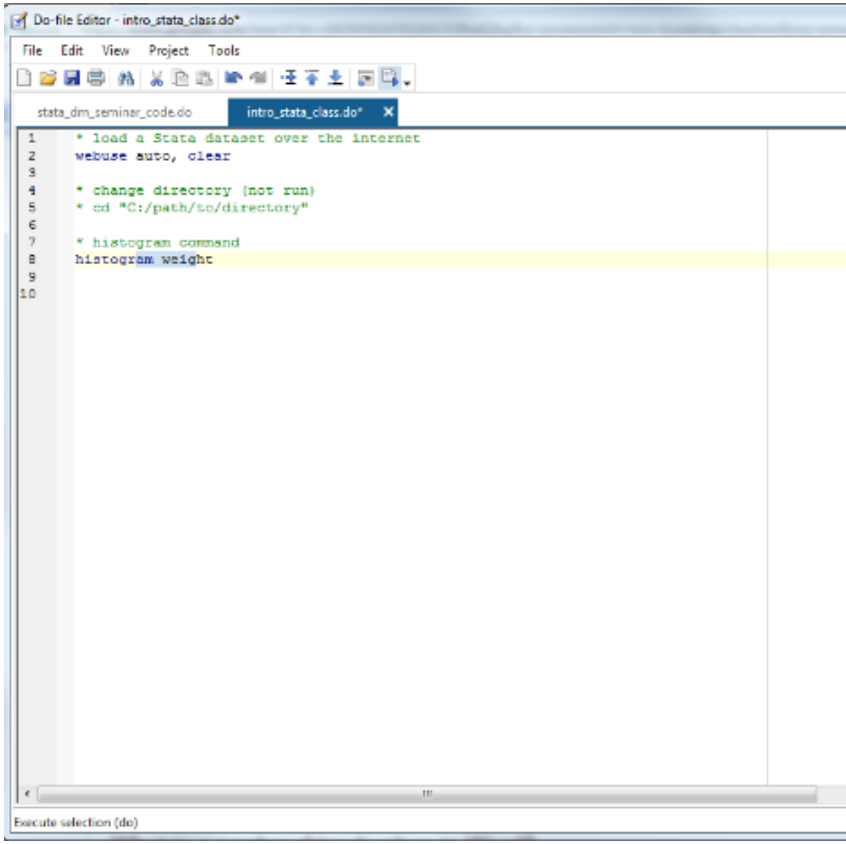




The do-file editor is a text file editor specialized for Stata

# Running commands from the do-file

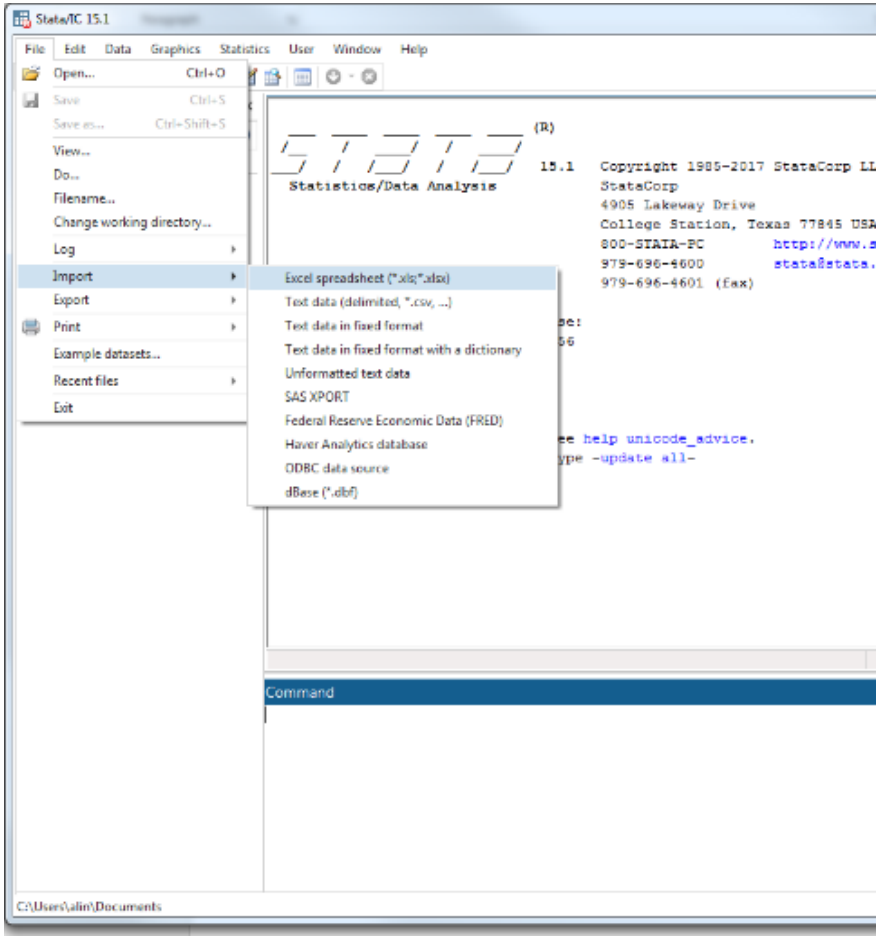
The do-file editor colors Stata commands blue
Comments, which are not executed, are usually preceded by * and are colored green

To run a command from the do-file, highlight part or all of the command, and then hit the “Execute(do)” icon, the rightmost icon on the do-file editor toolbar
Multiple commands can be selected and executed



# Import data

Select File -> Import data with the data type

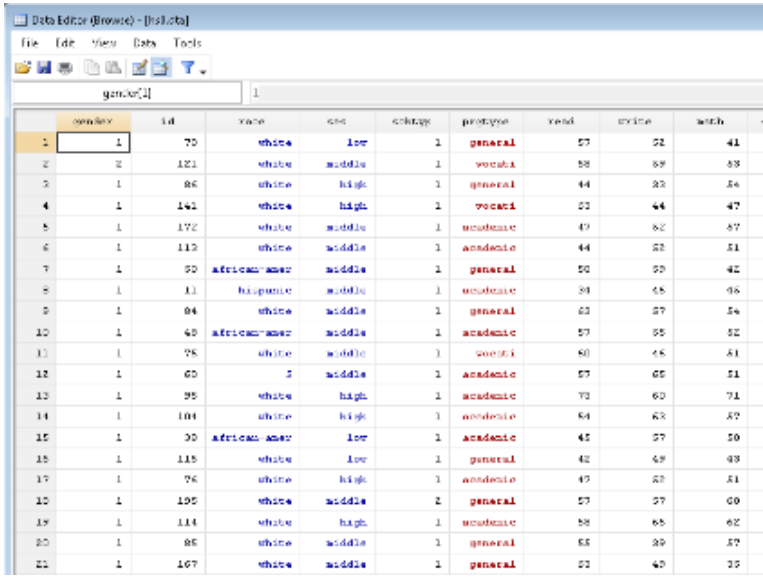


# Browsing the dataset

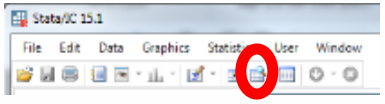
Once the data are loaded, we can view the dataset as a spreadsheet using the command

```
browse
```

The magnifying glass with spreadsheet icon also browses the dataset




Black columns are numeric, red columns are strings, and blue columns are numeric with string labels



# Summarizing continuous variables

The `summarize` command calculates a variable's:

- number of non-missing observations
- mean
- standard deviation
- min and max

```
* summarize continuous variables
summarize read math

    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+--------------------------------------------------------
        read |       200       52.23    10.25294         28         76
        math |       200      52.645    9.368448         33         75

* summarize read and math for females
summarize read math if gender == 2
    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+--------------------------------------------------------
        read |       109    51.73394    10.05783         28         76
        math |       109     52.3945    9.151015         33         72
```

# Detailed summaries

Use the `detail` option with `summary` to get more estimates that characterize the distribution, such as:

- percentiles (including the median at 50th percentile)
- variance
- skewness
- kurtosis

```
* detailed summary of read for females
summarize read if gender == 2, detail
```

```
                    reading score
-------------------------------------------------------------
      Percentiles      Smallest
 1%           34             28
 5%           36             34
10%           39             34       Obs                 109
25%           44             35       Sum of Wgt.         109

50%           50                      Mean           51.73394
                        Largest       Std. Dev.      10.05783
75%           57             71
90%           68             73       Variance        101.16
95%           68             73       Skewness       .3234174
99%           73             76       Kurtosis       2.500028
```

# Frequencies of categorical variables

`tabulate` displays counts of each value of a variable

useful for variables with a limited number of levels

use the `nolabel` option to display the underlying numeric values (by removing value labels)

```
* tabulate frequencies of ses
tabulate ses
        ses |      Freq.     Percent        Cum.
------------+-----------------------------------
        low |         47       23.50       23.50
     middle |         95       47.50       71.00
       high |         58       29.00      100.00
------------+-----------------------------------
      Total |        200      100.00

* remove labels
tab ses, nolabel
        ses |      Freq.     Percent        Cum.
------------+-----------------------------------
          1 |         47       23.50       23.50
          2 |         95       47.50       71.00
          3 |         58       29.00      100.00
------------+-----------------------------------
      Total |        200      100.00
```

# Two-way tabulations

`tabulate` can also calculate the joint frequencies of two variables
Use the `row` and `col` options to display row and column percentages
We may have found an error in a race value (5?)

```
* with row percentages
tab race ses, row

|                                  ses
       race   |      low     middle       high |     Total
-------------+---------------------------------+----------
   hispanic |        9         11          4 |        24
            |    37.50      45.83      16.67 |    100.00
-------------+---------------------------------+----------
      asian |        3          5          3 |        11
            |    27.27      45.45      27.27 |    100.00
-------------+---------------------------------+----------
african-amer |       11          6          3 |        20
            |    55.00      30.00      15.00 |    100.00
-------------+---------------------------------+----------
      white |       24         71         48 |       143
            |    16.78      49.65      33.57 |    100.00
-------------+---------------------------------+----------
          5 |        0          2          0 |         2
            |     0.00     100.00       0.00 |    100.00
-------------+---------------------------------+----------
      Total |       47         95         58 |       200
            |    23.50      47.50      29.00 |    100.00
```

# Replacing values

Use `replace` to replace values of existing variables

Often used with `if` to replace values for a subset of observations

Here we see the use of the missing numeric value indicator **.**

Missing value for strings is “”

```
* replace total with just (math+socst)
*  if science is missing
replace total = math + science if science == .

* no missing totals now
summarize total
```

```
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
       total |        200      155.42    25.47565         74        213
```

# Creating dummy indicators

It is often necessary to create variables that are 0/1 indicators for belonging to a category of another variable, where 0=FALSE and 1=TRUE

often called dummy variables or indicators

Remember that Stata often prefers to work with numeric variables

```
* create a variable that equals 1 if prgtype
*  equals academic, 0 otherwise
gen academic = 0
replace academic = 1 if prgtype == "academic"
tab prgtype academic
```

```
           |      academic
   prgtype |         0         1 |     Total
-----------+----------------------+----------
  academic |         0       105 |       105
   general |        45         0 |        45
    vocati |        50         0 |        50
-----------+----------------------+----------
     Total |        95       105 |       200
```

# egen

`egen` (extended generate) creates variables using a wide array of functions, which include:

1. statistical functions that accept multiple variables as arguments
   e.g. means across several variables
2. functions that accept a single variable, but do not involve simple arithmetic operations
   e.g. standardizing a variable (subtract mean and divide by standard deviation)

See the help file for `egen` to see a full list of available functions

```
* egen to generate variables with functions
*  rowmean returns mean of all non-missing values
egen meantest = rowmean(read math science socst)
summarize meantest read math science socst
```

| Variable | Obs | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| meantest | 200 | 52.28042 | 8.400239 | 32.5 | 70.66666 |
| read | 200 | 52.23 | 10.25294 | 28 | 76 |
| math | 200 | 52.645 | 9.368448 | 33 | 75 |
| science | 195 | 51.66154 | 9.866026 | 26 | 74 |
| socst | 200 | 52.405 | 10.73579 | 26 | 71 |

# Download entire database

Search All 11/974 | Identifying Information 1/8 | Identifying Information, cont. 1/34 | Company

**Select** All

sic

- Reversal - Restructuring/Acq Basic EPS Effect (rreps)
- Acquisition/Merger Basic EPS Effect (aqeps)
- As Reported Core - Basic EPS Effect (arceeps)
- Extinguishment of Debt Basic EPS Effect (dteeps)
- Earnings Per Share (Basic) - Including Extraordinary Items (epspi)
- Earnings Per Share (Basic) - Excluding Extraordinary Items (epspx)
- Impairments of Goodwill Basic EPS Effect (gdwlieps)
- Gain/Loss on Sale (Core Earnings Adjusted) Basic

**Selected** Clear All (11)

- CUSIP (cusip)
- Current Assets - Total (act)
- Assets - Total (at)
- Current Liabilities - Total (lct)
- Liabilities - Total (lt)
- Net Income (Loss) (ni)
- Auditors'' Remuneraton (rmum)
- Cash and Cash Equivalents - Increase/(Decrease) (chech)
- Auditor (au)
- Market Value - Total - Fiscal (mkvalt)
- Standard Industry Classification Code (sic)

# STATA Sample code

```
use "/Users/Downloads/crbncgzhzmrvg9dg.dta", clear

* 2-digit industry code
gen industry=substr(sic,1,2)

* control for fixed effect
egen long FE2=group(industry fyear)

gen Lev=lt/at
gen Size=ln(mkvalt)
gen Profit=ni/at
gen Cash=chech/at

* sort data
sort cusip fyear
winsor2 Profit Lev Size Cash, replace cuts(1 99) by(FE2)
by cusip: gen Profit1 = Profit[_n+1]

sum Profit Lev Size Cash
cor Profit1 Profit Lev Size Cash
```

# Main regression

```
areg Profit1 Profit, a(industry)
est store d1
areg Profit1 Lev Size Cash, a(industry)
est store d2
areg Profit1 Profit Lev Size Cash, a(industry)
est store d3
areg Profit1 Profit, a(fyear)
est store d4
areg Profit1 Lev Size Cash, a(fyear)
est store d5
areg Profit1 Profit Lev Size Cash, a(fyear)
est store d6
areg Profit1 Profit, a(FE2)
est store d7
areg Profit1 Lev Size Cash, a(FE2)
est store d8
areg Profit1 Profit Lev Size Cash, a(FE2)
est store d9
esttab d1 d2 d3 d4 d5 d6 d7 d8 d9 using maintable.csv, nogaps replace t b(3) ar2 star (*  0.10
** 0.05   *** 0.01)
```

# Create subsample by group

```
bysort FE2: egen size_median=median(Size)
gen small=.
replace small=0 if Size<size_median
replace small=1 if Size>size_median

bysort FE2: egen risk_median=median(Lev)
gen risky=.
replace risky=0 if Lev<risk_median
replace risky=1 if Lev>risk_median
```

# Subsample Analysis

```
areg Profit1 Profit if small==1, a(FE2)
est store a1
areg Profit1 Lev Size Cash if small==1, a(FE2)
est store a2
areg Profit1 Profit Lev Size Cash if small==1, a(FE2)
est store a3
areg Profit1 Profit if small==0, a(FE2)
est store a4
areg Profit1 Lev Size Cash if small==0, a(FE2)
est store a5
areg Profit1 Profit Lev Size Cash if small==0, a(FE2)
est store a6

esttab a1 a2 a3 a4 a5 a6 using sub_size.csv, nogaps replace t b(3) ar2 star (*  0.10
** 0.05   *** 0.01)
```

# Subsample Analysis

```
areg Profit1 Profit if risky==1, a(FE2)
est store b1
areg Profit1 Lev Size Cash if risky==1, a(FE2)
est store b2
areg Profit1 Profit Lev Size Cash if risky==1, a(FE2)
est store b3
areg Profit1 Profit if risky==0, a(FE2)
est store b4
areg Profit1 Lev Size Cash if risky==0, a(FE2)
est store b5
areg Profit1 Profit Lev Size Cash if risky==0, a(FE2)
est store b6
esttab a1 a2 a3 a4 a5 a6 using sub_risk.csv, nogaps
replace t b(3) ar2 star (*  0.10  ** 0.05   *** 0.01)
```

# Event study

```
gen COVID=0
replace COVID=1 if fyear>2019 & fyear<2022

gen ProfitxCOVID=Profit*COVID

areg Profit1 Profit COVID ProfitxCOVID, a(FE2)
est store d7
areg Profit1 COVID Lev Size Cash, a(FE2)
est store d8
areg Profit1 COVID ProfitxCOVID Profit Lev Size Cash,
a(FE2)
est store d9
esttab d1 d2 d3 d4 d5 d6 d7 d8 d9 using additon1.csv,
nogaps replace t b(3) ar2 star (*  0.10  ** 0.05   ***
0.01)
```

# Seminar Activity

- Save your data into .dta
- Find the summary and correlation matrix of the main variables, and run at least three regression models
- Do at least one subsample analysis for the same three regression models
- Identify an event (e.g. GFC, COVID etc) and do an event study